

Profitable Flight-Delay Prediction Under EC261 - A Diagnosed Negative Result

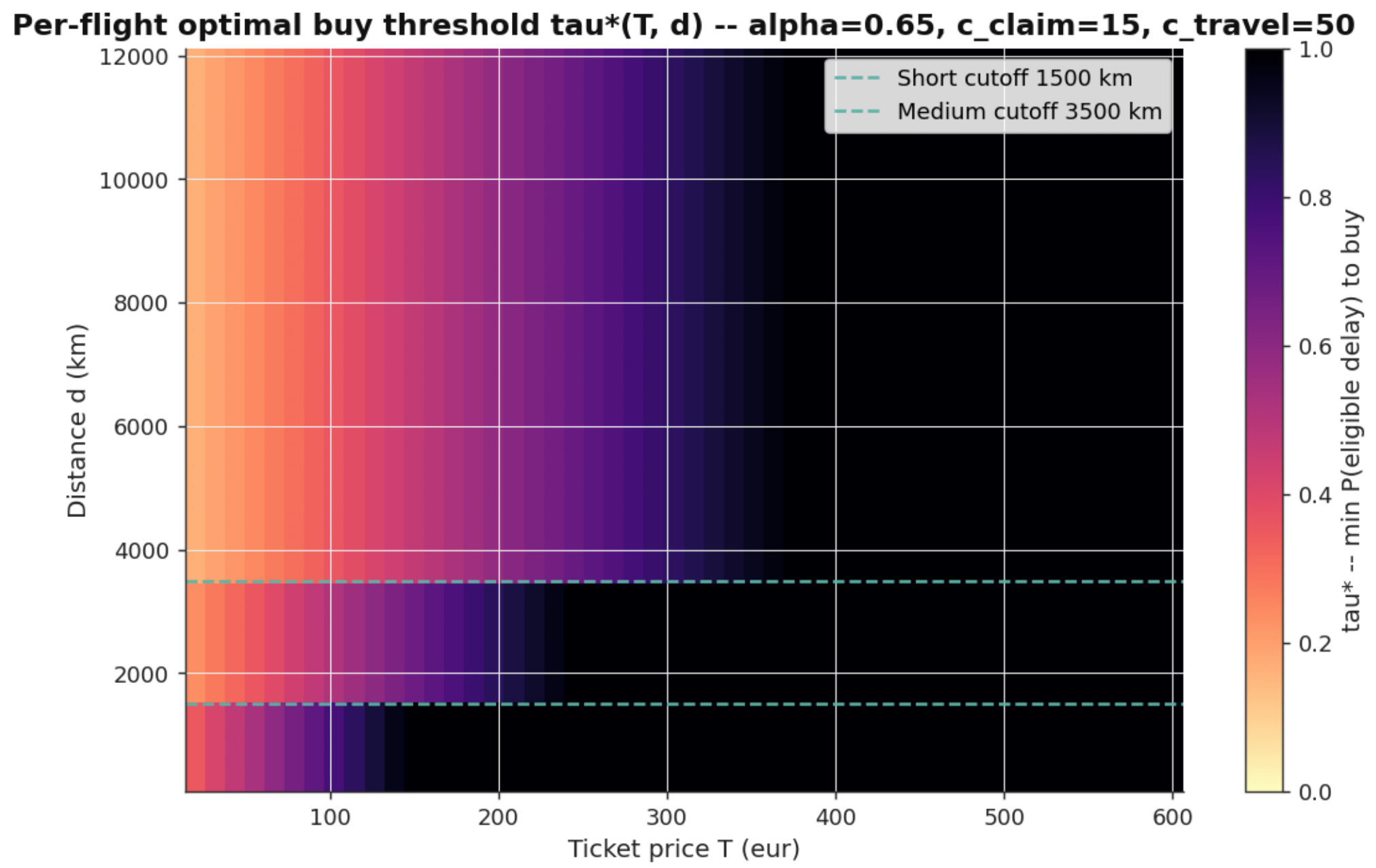
Rahal · Arida · Khoury · Moucattash · Nsour · Albilleh · IE University · BCSAI2025CSAI.2.M.A C2

No positive-EV ticket exists: every calibrated model abstains (ROI 0%). Required confidence 63% is $8\times$ any achievable delay rate: a 3.89M-flight EUROCONTROL test confirms it ($\rho = -1.00$).

MOTIVATION

EC261 pays EUR 250-600 for carrier 3h+ delays. Cheap ticket + likely delay = positive-EV bet?

ML as a decision, not accuracy.

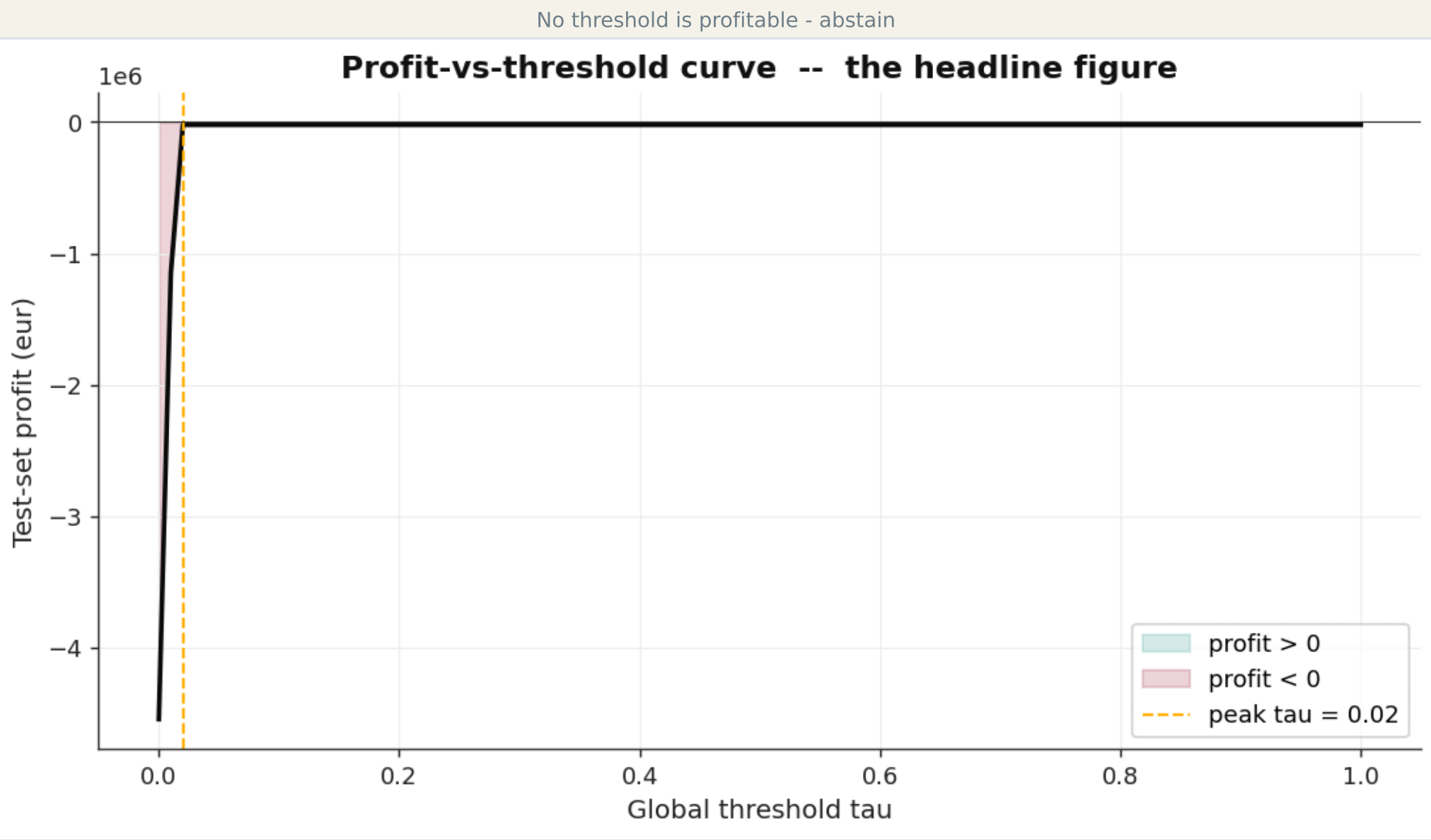


DATA

Real US BTS 2024: 6.97 M -> seeded 150 k. EC261 base rate 1.18%. EUROCONTROL 3.89 M real flights (transfer).

METHOD

Leakage-audited pipeline. 6-model ladder, isotonic calibration. Grid/Random/Bayes on a custom profit scorer.

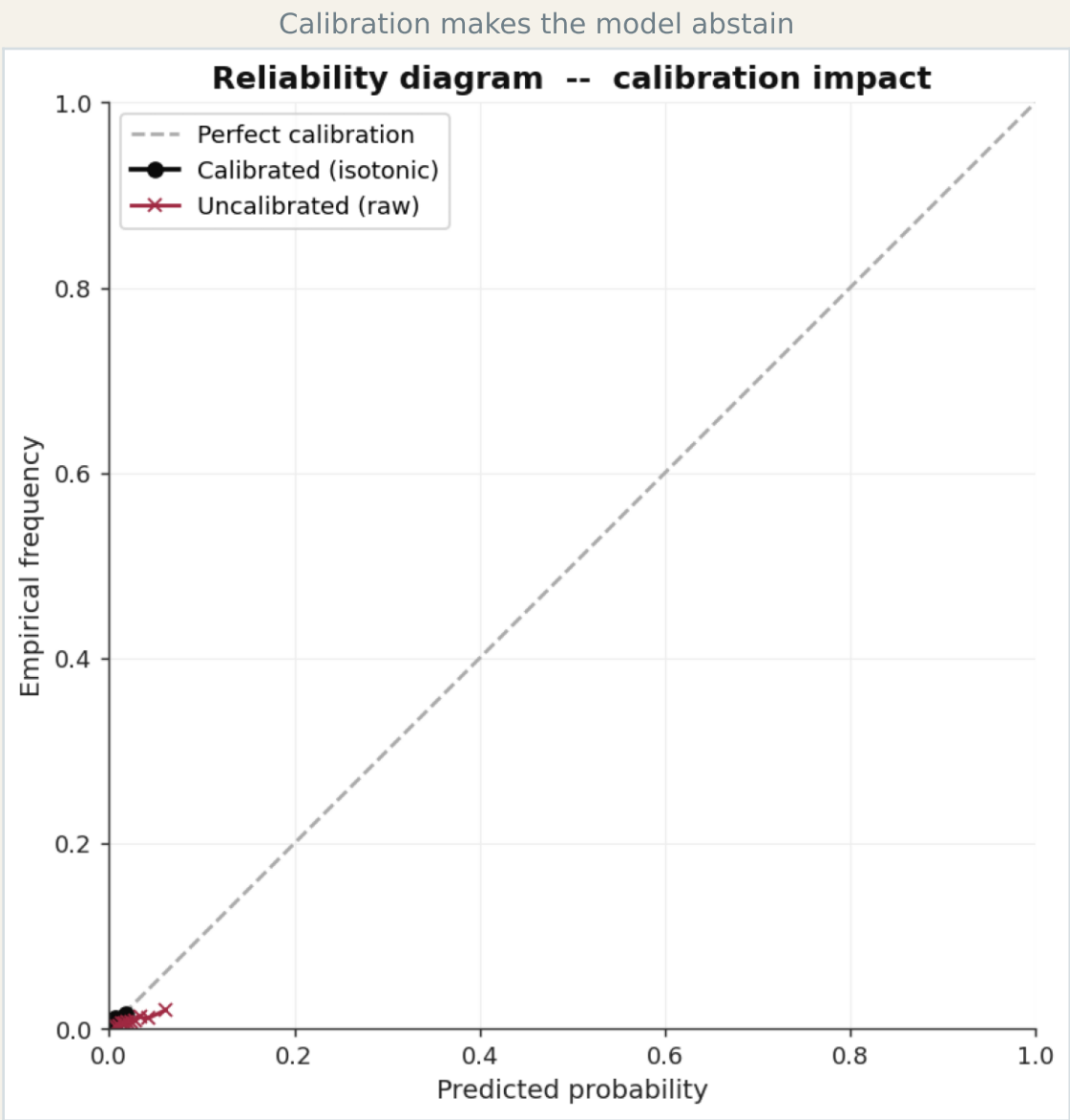


EU TRANSFER

Spearman $\rho = -1.00$ (ranking inverted). Top-k lift ≤ 1.03 . EU signal 15.6->39.2% by haul - unusable.

RESULTS (test n=29,604)

All calibrated models: ROI 0%, 0 buys. Best RF+iso ROC 0.65. Uncal XGB -191.7% -> calibration decisive.



WHY / REFS

EC261 cap $\times \alpha 0.65 <$ ticket + EUR 65 for a ~1% event. Structural. Negative result = finding (brief 1/2).